Monday Warm Up: Unit 2, DC Circuits

Prof. Jordan C. Hanson

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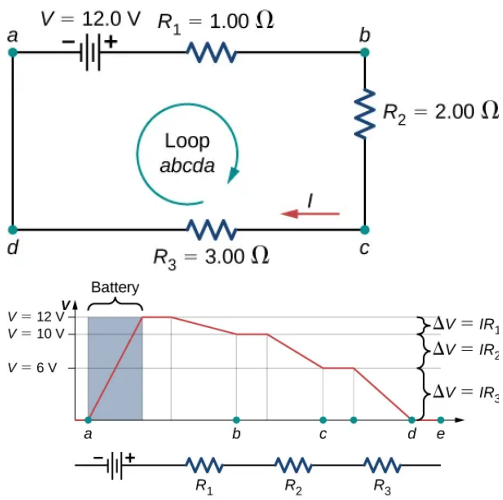

Figure 1: (Top) Three resistors in series with a battery emf. (Bottom) Chart of the voltage as a function of location around the series loop.

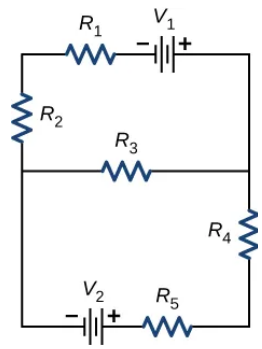

Figure 2: A circuit with two batteries and five resistors.

1 Memory Bank

- $V = IR$ ... The macroscopic version of Ohm's law
- $R_{\text{tot}} = R_1 + R_2$ ... Resistors in series.
- $R_{\text{tot}}^{-1} = R_1^{-1} + R_2^{-1}$ ... Resistors in parallel.
- $V_{\text{terminal}} = \mathcal{E} - Ir$ ... The terminal voltage on a battery with emf $\mathcal{E}$, current I, and *internal resistance* r.
- $P = IV = I^2R = V^2/R$ ... Expressions for the power in a DC circuit.
- $(R_1 \pm \sigma_1) + (R_2 \pm \sigma_2) = (R_1 + R_2) \pm \sqrt{\sigma_1^2 + \sigma_2^2}$ Adding two measurements with associated errors.

2 Kirchhoff's Rules

1. Consider Fig. 1 (top). (a) What current flows from the battery? (b) What is the total power consumption?

2. Consider Fig. 1 (bottom). Imagine moving along the path from point a to point d, and then back to a. According to the chart in Fig. 1 (bottom), what is the total change in voltage?

3. The loop technique in the previous exercise technically works for any loop within a circuit (conservation of energy). Consider Fig. 2, and let I_3 be the current through R_3, I_4 be the current through R_4, and I_1 be the current through R_1. (a) Use the loop technique to prove that

$$V_1 - I_3R_3 - I_1R_2 - I_1R_1 = 0 \tag{1}$$

(b) Use a loop involving V_2 to create a similar equation as found in part (a) for the lower loop.

4. The *junction technique* states that (by charge conservation) the currents leaving a junction should sum to the currents entering a junction. Using the loop and the junction techniques, solve for all currents through all resistors in Fig. 2.